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**EX NO. 3**

**ASSET MANAGEMENT USING HYPERLEDGER FABRIC CHAINCODE AIM**

To perform asset management operations such as creating, reading, updating, transferring, and deleting assets using JavaScript chaincode in a Hyperledger Fabric network.

**SOFTWARE REQUIREMENTS**

- Ubuntu / Kali Linux
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Curl
- jq
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images

**HARDWARE REQUIREMENTS**

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 20 GB free disk space
- **Internet**

**Connection PROCEDURE**

**Step 1: Navigate to the Hyperledger Fabric Test**

**Network** Open the terminal and navigate to the

Fabric test network. cd ~/fabric-dev/fabric-

samples/test-network

**Step 2: Stop Any Existing Fabric Network**

To ensure that no previous network is running, execute:

./network.sh down

**Step 3: Start the Fabric Network**

Start the Fabric test network with Certificate Authorities and create the application channel.

./network.sh up createChannel -ca

The command performs the following operations:

- Starts the Orderer.
- Starts Org1 and Org2 peers.
- Starts Certificate Authorities.
- Generates cryptographic materials.
- Creates the application channel.

**Step 4: Deploy the Asset Transfer Chaincode**

Deploy the JavaScript Asset Transfer Basic chaincode.

./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript

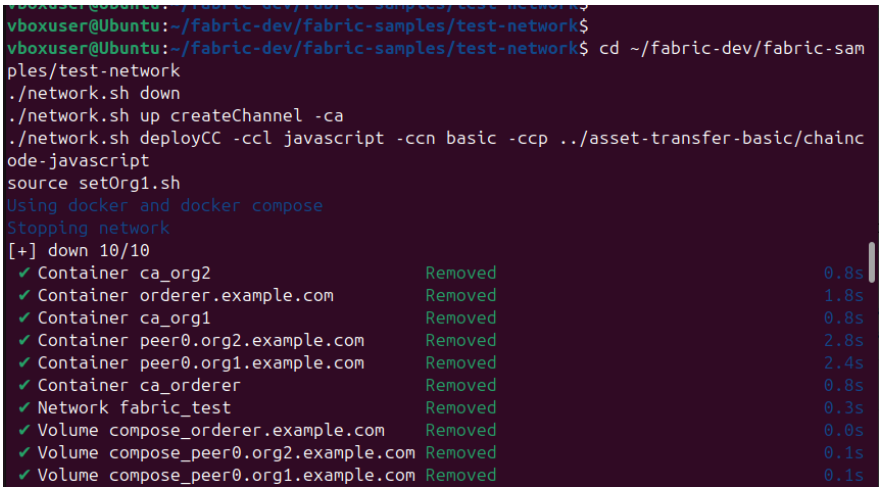
The command performs:

- Chaincode packaging
- Chaincode installation
- Chaincode approval
- Chaincode commitment

**Step 5: Configure Org1 Peer Environment**

Load the Org1 peer

environment. source setOrg1.sh



If the helper script is not available, use:

export PATH=${PWD}/../bin:$PATH

export

FABRIC_CFG_PATH=${PWD}/../config/

export CORE_PEER_TLS_ENABLED=true

export

CORE_PEER_LOCALMSPID="Org1MSP"

export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.exa mple.com/peers/peer0.org1.example.com/tls/ca.crt

export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example
.com/users/Admin@org1.example.com/

msp export

CORE_PEER_ADDRESS=localhost:7051

export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer
.example.com/msp/tlscacerts/tlsca.example.com-cert.pem

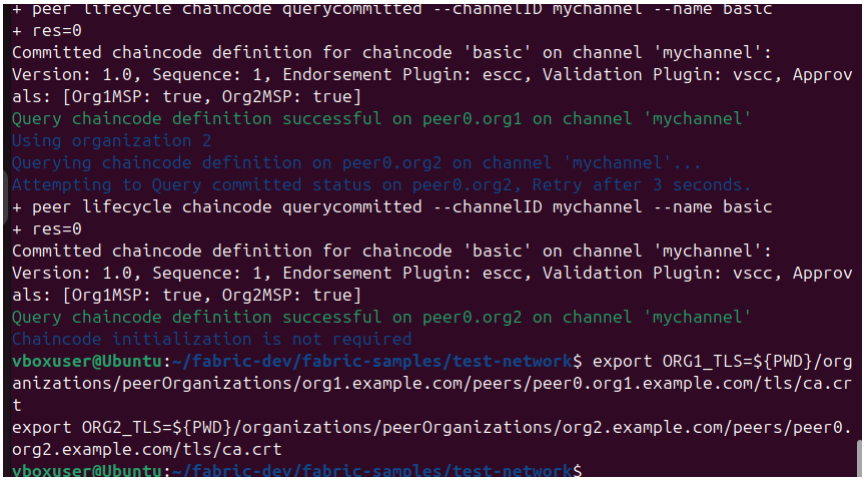
**Step 6: Set TLS Certificate Paths for Both Organizations**

Set the TLS certificate paths for Org1 and

Org2. export
ORG1_TLS=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1. example.com/tls/ca.crt

export
ORG2_TLS=${PWD}/organizations/peerOrganizations/org2.example.com/peers/peer0.org2. example.com/tls/ca.crt

These certificates are required because the default channel endorsement policy requires transactions to be endorsed by both organizations.


+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approv
als: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approv
als: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ export ORG1_TLS=${PWD}/org
anizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.cr
t
export ORG2_TLS=${PWD}/organizations/peerOrganizations/org2.example.com/peers/peer0.
org2.example.com/tls/ca.crt
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$

**ASSET MANAGEMENT OPERATIONS**

**Step 7: Initialize the Ledger**

Initialize the ledger with the predefined assets.

**Run this command as a single line:**

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"InitLedger","Args":[]}'

**Step 8: Query All Existing Assets**

Display all assets currently stored in the ledger.

peer chaincode query -C mychannel -n basic -c

'{"Args":["GetAllAssets"]}' The command displays the existing asset

records.

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o l
ocalhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDER
ER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_
TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"Ini
tLedger","Args":[]}'
2026-08-28 18:49:27.737 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chain
code invoke successful. result: status:200
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C my
channel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docTy
pe":"asset"},{"AppraisedValue":900,"Color":"White","ID":"asset11","Owner":"Arun","Si
ze":12},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"d
ocType":"asset"},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin So
o","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yellow","ID":"asset4"
,"Owner":"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","I
D":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":800,"Co
lor":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$
```

**Step 9: Create a New Asset**

Create a new asset named asset11.

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"CreateAsset","Args":["asset11","White","12","Arun","900"]}'

The asset contains the following information:

- **Asset ID:** asset11
- **Color:** White

- **Size:** 12
- **Owner:** Arun
- **Appraised Value:** 900

**Step 10: Read the Created Asset**

Verify that asset11 was successfully created.

peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'

**Expected Output:**

{

"ID": "asset11",

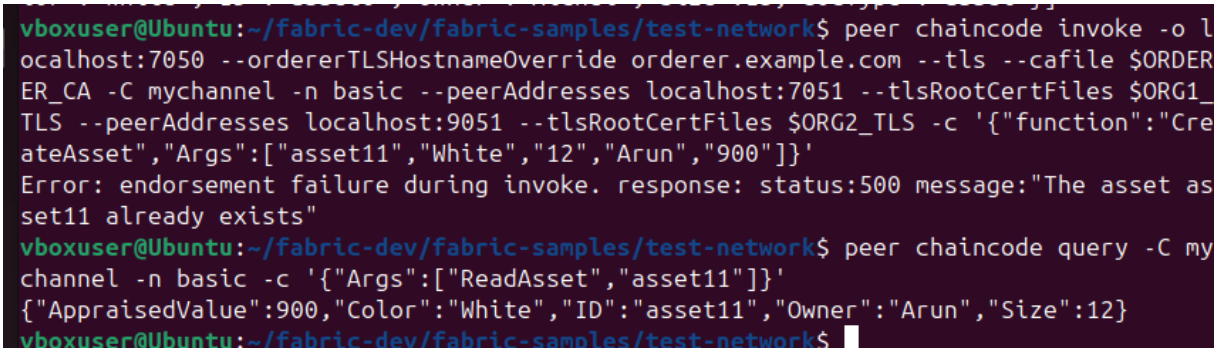
"Color": "White",

"Size": 12,

"Owner": "Arun",

"AppraisedValue": 900

}




**Step 11: Update the Asset**

Update the details of

asset11. The new values

are:

- **Color:** Black
- **Size:** 15
- **Owner:** Arun
- **Appraised Value:**

1200 Run:

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"UpdateAsset","Args":["asset11","Black","15","Arun","1200"]}'

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o l
ocalhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDER
ER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_
TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"Upd
ateAsset","Args":["asset11","Black","15","Arun","1200"]}'
2026-08-28 18:51:13.344 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chain
code invoke successful. result: status:200 payload:"{\"type\":\"Buffer\",\"data\":[]
}"
```

**Step 12: Verify the Updated Asset**

Query asset11 again.

peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'

**Expected Output:**

{

"ID": "asset11",

"Color": "Black",

"Size": 15,

"Owner": "Arun",

"AppraisedValue": 1200

}

**Step 13: Transfer Asset Ownership**

Transfer ownership of asset11 from Arun to David.

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"TransferAsset","Args":["asset11","David"]}'

**Expected Output:**

Chaincode invoke successful

**Step 14: Verify the New Owner**

Query asset11 to verify that ownership has changed.

peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'

**Expected Output:**

{

"ID": "asset11",

"Color": "Black",

"Size": 15,

"Owner":

"David",

"AppraisedValue": 1200

}

The ownership of asset11 has been successfully transferred from **Arun** to **David**.


**Step 15: Delete the Asset**

Delete asset11 from the ledger.

peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $ORG1_TLS --peerAddresses localhost:9051 --tlsRootCertFiles $ORG2_TLS -c '{"function":"DeleteAsset","Args":["asset11"]}'

**Expected Output:**

Chaincode invoke successful

**Step 16: Verify Asset Deletion**

Try to read asset11 after deletion.

peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset11"]}'

This confirms that the asset was successfully deleted from the ledger.


**Step 17: Stop the Fabric Network**

After completing all asset management operations, stop the Fabric network.

./network.sh down

**FLOW OF THE EXPERIMENT**

Start Fabric Network

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Create Application Channel

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Deploy Chaincode

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Initialize

Ledger

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▼

Create

Asset

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▼

ReadAsset

▼

Update

Asset

|

▼

Read Updated Asset

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▼

Transfer Ownership

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▼

Verify New Owner

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▼

Delete

Asset

|

▼

Verify Deletion

|

▼

Stop Network

**RESULT**

The asset management operations were successfully performed using Hyperledger Fabric JavaScript chaincode. A new asset named **asset11** was created with the specified details, successfully read from the ledger, updated, and transferred from **Arun** to **David**. Finally, the asset was deleted and its absence was verified. Thus, the complete asset lifecycle operations were successfully implemented and executed on the Hyperledger Fabric network.